

1 Cell structure

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Light microscopes resolve cells and tissues; electron microscopes resolve organelles and membranes. Magnification = image size/actual size, with matching units. Prokaryotes lack a membrane-bound nucleus and 80S ribosomes; eukaryotic organelles compartmentalise metabolic reactions. Resolution, unlike magnification, determines the smallest separable detail.

Worked example: follow the reasoning

A cell 25 μm long is drawn as 50 mm: $50\,000\,\mu\text{m}/25\,\mu\text{m} = \times 2000$ magnification.

Quick check · multiple choice

A What determines the smallest resolvable detail in a micrograph?

[1]

- A Resolution
- B Image colour
- C Printed page size
- D Number of labels

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A micrograph shows a 12 mm chloroplast that is actually 6 μm long. Calculate magnification and name one organelle present in plant cells but absent from typical animal cells.

[3]

Final answer: _____

Answers and explanation

Quick check: A. Resolution

Question 1 [3 marks]

12 mm = 12 000 μm ; magnification = 2000 [2]. Chloroplast or large permanent vacuole [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.